

ASSIGNMENT - 9**Complex form of Fourier series**

1. Find the complex Fourier series of $f(x) = x, -\pi < x < \pi$ integer.

$$\text{Ans: } f(x) = i \sum_{-\infty}^{\infty} \frac{(-1)^n}{n} e^{inx}$$

2. Find the complex form of the Fourier series of the $f(x) = e^x$, in $-\pi \leq x \leq \pi$.

$$\text{Ans: } f(x) = \frac{\sinh \pi}{\pi} \sum_{-\infty}^{\infty} \frac{(1+in)(-1)^n}{1+n^2} e^{inx}$$